

AOMORI, Japan

WMO: 475750

Lat: 40.8219N

Lon: 140.7689E

Elev: 4

StdP: 101.28

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.6	-5.6	-10.2	1.6	-3.2	-9.1	1.7	-2.5	11.2	-0.8	10.2	-1.1	4.6	210	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.9	30.6	23.9	29.0	23.2	27.5	22.6	24.9	29.0	24.2	27.8	23.5	26.7	5.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	18.5	27.2	23.0	17.8	26.6	22.3	17.0	26.0	76.0	29.1	73.1	28.0	70.3	26.8	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.4	9.2	8.2	-8.6	33.5	1.3	1.8	-9.6	34.8	-10.3	35.9	-11.1	36.9	-12.1	38.1		
			-9.2	25.9	1.3	0.9	-10.1	26.5	-10.8	27.0	-11.5	27.5	-12.4	28.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	-1.1	-0.5	2.8	8.7	14.0	17.8	22.2	23.9	20.1	13.7	7.2	1.4	(6)
(7)		DBStd	9.11	2.40	2.72	3.12	3.36	2.85	2.51	2.81	2.54	2.96	2.97	3.77	3.00	(7)
(8)		HDD10.0	1295	343	293	224	63	3	0	0	0	4	99	266		(8)
(9)		HDD18.3	3095	602	526	482	289	138	38	2	0	14	146	335	523	(9)
(10)		CDD10.0	1633	0	0	1	24	127	235	379	430	303	119	14	0	(10)
(11)		CDD18.3	392	0	0	0	1	3	23	123	172	67	2	0	0	(11)
(12)		CDH23.3	2186	0	0	0	4	26	83	653	1113	302	4	0	0	(12)
(13)		CDH26.7	509	0	0	0	0	3	13	154	281	58	0	0	0	(13)
(14)	Wind	WSAvg	3.7	4.2	4.2	4.4	4.2	3.7	3.2	3.1	3.0	3.1	3.4	3.8	4.1	(14)
(15)	Precipitation	PrecAvg	1360	146	101	75	68	79	75	130	141	132	117	139	156	(15)
(16)		PrecMax	1666	278	187	181	143	170	164	264	317	345	241	317	240	(16)
(17)		PrecMin	1007	63	32	17	14	37	9	21	26	39	44	63	93	(17)
(18)		PrecStd	202	45	40	35	30	34	37	66	82	68	55	54	45	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.7	9.4	15.9	23.2	26.4	28.6	32.3	33.3	30.9	23.5	19.2	12.3	(19)
(20)			MCWB	4.1	5.8	9.2	13.3	15.7	19.6	24.3	24.8	24.2	17.3	13.3	9.3	(20)
(21)		2%	DB	4.9	6.6	12.1	19.1	23.1	25.6	30.2	31.1	28.3	21.8	16.7	9.7	(21)
(22)			MCWB	2.5	3.5	7.2	11.1	14.7	18.9	23.5	24.5	22.6	16.5	12.3	6.9	(22)
(23)		5%	DB	3.7	5.2	9.8	16.5	20.9	23.7	28.3	29.5	26.3	20.4	14.9	7.7	(23)
(24)			MCWB	1.7	2.5	5.9	9.9	13.8	18.2	23.1	23.9	21.3	15.7	11.2	5.3	(24)
(25)		10%	DB	2.6	3.9	7.8	14.4	19.2	22.2	26.7	28.1	24.9	19.1	13.1	6.0	(25)
(26)			MCWB	0.8	1.5	4.4	9.1	13.5	17.6	22.3	23.3	20.3	14.8	10.2	3.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.6	6.3	10.0	14.5	17.6	21.4	25.2	26.0	25.0	19.7	14.7	9.9	(27)
(28)			MCDWB	6.2	8.7	14.1	19.5	22.6	25.4	30.1	30.4	28.9	22.0	17.2	11.9	(28)
(29)		2%	WB	3.0	4.1	7.9	12.4	16.2	20.1	24.4	25.3	23.6	17.7	13.0	7.5	(29)
(30)			MCDWB	4.6	6.0	11.5	17.0	20.7	23.7	28.7	29.4	27.1	20.4	15.8	9.3	(30)
(31)		5%	WB	1.9	2.8	6.2	11.0	15.2	19.1	23.7	24.6	22.4	16.6	11.8	5.6	(31)
(32)			MCDWB	3.4	4.7	9.2	14.8	19.3	22.5	27.4	28.4	25.4	19.4	14.3	7.4	(32)
(33)		10%	WB	1.0	1.7	4.9	9.8	14.3	18.3	22.9	24.0	21.2	15.5	10.4	3.9	(33)
(34)			MCDWB	2.3	3.5	7.5	13.7	18.0	21.3	26.0	27.2	24.0	18.3	12.8	5.8	(34)
(35)	Mean Daily Temperature Range	MDBR	4.7	5.5	6.9	8.7	8.5	7.2	6.6	6.9	7.9	8.5	7.1	5.3	(35)	
(36)		5% DB	MCDBR	6.5	7.8	10.3	12.2	11.9	9.6	8.7	8.6	9.1	10.0	10.3	8.4	(36)
(37)			MCWBR	5.1	5.8	6.9	6.8	5.9	4.7	3.8	4.0	4.7	6.1	7.0	6.5	(37)
(38)		5% WB	MCDBR	6.5	7.9	9.8	10.3	9.8	7.9	7.4	7.4	8.1	8.4	9.8	8.4	(38)
(39)			MCWBR	5.5	6.3	7.2	6.7	5.6	4.2	3.5	3.6	4.6	5.8	7.4	7.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.377	0.421	0.493	0.542	0.556	0.531	0.534	0.500	0.441	0.420	0.408	0.391	(40)	
(41)		taud	2.006	1.899	1.826	1.784	1.831	1.972	2.034	2.140	2.259	2.183	2.108	2.038	(41)	
(42)		Ebn at Noon	728	753	744	741	744	764	757	772	795	757	695	675	(42)	
(43)		Edh at Noon	121	157	190	212	207	180	168	147	122	117	110	109	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.29	2.03	3.21	4.43	4.96	4.96	4.37	4.31	3.69	2.77	1.66	1.14	(44)	
(45)		RadStd	0.08	0.21	0.27	0.45	0.50	0.41	0.45	0.44	0.29	0.21	0.15	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page